of UWL and is often more pronounced in men. More intense forms of mental illness such as paranoid disorders may lead to delusions about food and cause weight loss. Alcoholism can be a significant source of weight loss and malnutrition.

Elderly persons living in poverty may have to choose whether to purchase food or use the money for other expenses, including medications. Screening questions can probe whether patients have run out of food or whether they routinely purchase less than they need. Institutionalization is an independent risk factor, as up to 30–50% of nursing home patients have inadequate food intake.

Medications can cause anorexia, nausea, vomiting, gastrointestinal distress, diarrhea, dry mouth, and changes in taste. This is particularly an issue in the elderly, many of whom take five or more medications.

ASSESSMENT

The four major manifestations of UWL are (1) anorexia (loss of appetite), (2) sarcopenia (loss of muscle mass), (3) cachexia (a syndrome that combines weight loss, loss of muscle and adipose tissue, anorexia, and weakness), and (4) dehydration. The current obesity epidemic adds complexity, as excess adipose tissue can mask the development of sarcopenia and delay awareness of the development of cachexia. If it is not possible to measure weight directly, a change in clothing size, corroboration of weight loss by a relative or friend, and a numeric estimate of weight loss provided by the patient are suggestive of true weight loss.

Initial assessment includes a comprehensive history and physical, a complete blood count, tests of liver enzyme levels, C-reactive protein, erythrocyte sedimentation rate, renal function studies, thyroid function tests, chest radiography, and an abdominal ultrasound **(Table 50-2)**. Age-, sex-, and risk factor–specific cancer screening tests, such as fecal occult blood, colonoscopy, or mammography, should be performed **(Chap. 75)**. Patients at risk should have HIV testing. All elderly patients with weight loss should undergo screening for dementia and depression by using instruments such as the Mini-Mental State Examination and the Geriatric Depression Scale, respectively **(Chap. 489)**. The Mini Nutritional Assessment (*www.mna-elderly.com*) and the Nutrition Screening Initiative (*http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1694757/*) are also available for the nutritional assessment of elderly patients. Almost all patients with a malignancy and >90% of those with other organic diseases have at least one laboratory abnormality. In patients presenting with substantial UWL, major organic and malignant diseases are unlikely when a baseline evaluation is completely normal. Careful follow-up rather than additional undirected testing is advised because the prognosis of weight loss of undetermined cause is generally favorable.

TABLE 50-2 Assessment and Testing for Involuntary Weight Loss

Indications	Laboratory
5% weight loss in 6 mo	Complete blood count
	Comprehensive electrolyte and metabolic panel, including liver and renal function tests
Body mass index <21	Thyroid function tests
25% of food left uneaten after 7 d	Erythrocyte sedimentation rate
Change in fit of clothing	C-reactive protein
Change in appetite, smell, or taste	Ferritin
Abdominal pain, nausea, vomiting, diarrhea, constipation, dysphagia	HIV testing, if indicated
Assessment	**Radiology**
Complete physical examination, including dental evaluation	Chest x-ray Abdominal ultrasound
Medication review	
Recommended cancer screening	
Mini-Mental State Examination[a]	
Mini-Nutritional Assessment[a]	
Nutrition Screening Initiative[a]	
Simplified Nutritional Assessment Questionnaire[a]	
Observation of eating[a]	
Activities of daily living[a]	
Instrumental activities of daily living[a]	

[a]May be more specific to assess weight loss in the elderly.

TREATMENT

Unintentional Weight Loss

The first priority in managing weight loss is to identify and treat the underlying causes. Treatment of underlying metabolic, social, psychiatric, dental, infectious, or other systemic disorders may be sufficient to restore weight and functional status gradually. Medications that cause nausea or anorexia should be withdrawn or changed, if possible. For those with unexplained UWL, oral nutritional supplements such as high-energy drinks sometimes reverse weight loss. Advising patients to choose appealing foods and to consume supplements between meals rather than with a meal may help minimize appetite suppression and facilitate increased overall intake. Orexigenic, anabolic, and anticytokine agents are not generally recommended. In selected patients, the antidepressant mirtazapine significantly increases body weight, body fat mass, and leptin concentration. However, side effects, including dizziness, fatigue, and somnolence, occur in about 10% of patients. Patients with wasting conditions who can comply with an appropriate exercise program gain muscle protein mass, strength, and endurance and may be more capable of performing ADLs.

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51 Gastrointestinal Bleeding

Loren Laine

Gastrointestinal bleeding (GIB) presents as either overt or occult bleeding. *Overt GIB* is manifested by *hematemesis*, vomitus of red blood or "coffee-grounds" material; *melena*, black, tarry stool; and/or *hematochezia*, passage of red or maroon blood from the rectum. In the absence of overt bleeding, *occult GIB* may present with *symptoms of blood loss or anemia* such as lightheadedness, syncope, angina, or dyspnea; with iron-deficiency anemia; or a positive fecal occult blood test on colorectal cancer screening. GIB is also categorized by the site of bleeding as upper, from the esophagus, stomach, or duodenum; lower, from the colon; small intestinal; or obscure GIB if the source is unclear.

GIB is the most common gastrointestinal condition leading to hospitalization in the United States, accounting for ~530,000 admissions annually and a case fatality of ~2%. Patients generally die from decompensation of other underlying illnesses rather than exsanguination.

SOURCES OF GASTROINTESTINAL BLEEDING

Upper Gastrointestinal Sources of Bleeding • **PEPTIC ULCERS** Peptic ulcers are the most common cause of upper GIB (UGIB), accounting for ~50% of UGIB hospitalizations. Features of an ulcer at endoscopy provide important prognostic information that guides subsequent management decisions **(Fig. 51-1)**. Approximately 20% of patients with bleeding ulcers have the highest-risk findings of active bleeding or a nonbleeding visible vessel; one-third of such patients have further bleeding that requires urgent surgery if they are treated conservatively. These patients benefit from endoscopic therapy such as bipolar electrocoagulation, heater probe, injection therapy (e.g., absolute alcohol, 1:10,000 epinephrine), clips, and/or topical hemostatic powder with reductions in bleeding, hospital stay, mortality, and costs. In contrast, patients with clean-based ulcers have rates of serious rebleeding approaching zero. If stable with no other reason for hospitalization, such patients may be discharged home after endoscopy.

Randomized controlled trials document that high-dose, proton pump inhibitor (PPI), given to reduce intragastric acid and thereby enhance clot stability, decreases further bleeding and mortality in patients with high-risk ulcers (active bleeding, nonbleeding visible vessel, adherent clot) when given after endoscopic therapy. Meta-analysis of randomized trials indicates that outcomes are comparable with high-dose PPIs given as a constant infusion or intermittently. Patients with lower-risk findings (flat pigmented spot, clean base) do not require endoscopic therapy and receive standard doses of oral PPI.

Approximately 10–50% of patients with bleeding ulcers rebleed within the next year if no preventive strategies are employed. Prevention of recurrent bleeding focuses on the three main factors in ulcer pathogenesis, *Helicobacter pylori*, nonsteroidal anti-inflammatory drugs (NSAIDs), and acid. Eradication of *H. pylori* in patients with bleeding ulcers decreases rebleeding rates to <5%. If a bleeding ulcer develops in a patient taking NSAIDs, the NSAIDs should be discontinued. If NSAIDs must be given, a cyclooxygenase-2 selective NSAID plus a PPI is recommended, based on results of a randomized trial. Patients with established cardiovascular disease who develop bleeding ulcers while taking low-dose aspirin for secondary prevention should not discontinue aspirin and, if aspirin is held, should restart aspirin once hemostasis is confirmed. A randomized trial showed that immediate reinstitution of aspirin was associated with a lower 8-week mortality compared to not restarting aspirin (1% vs 13%; hazard ratio, 0.2; 95% CI, 0.1–0.6). In contrast, aspirin should be discontinued in patients taking aspirin for primary prevention of cardiovascular events who develop UGIB. Patients with bleeding ulcers unrelated to *H. pylori* or NSAIDs should remain on PPI therapy indefinitely given a 42% incidence of rebleeding at 7 years without protective therapy. **Peptic ulcers are discussed in Chap. 335.**

MALLORY-WEISS TEARS Mallory-Weiss tears account for ~2–10% of UGIB hospitalizations. The classic history is vomiting, retching, or coughing preceding hematemesis. Bleeding from these tears, which are usually on the gastric side of the gastroesophageal junction, stops spontaneously in ~80–90% of patients and recurs in only 0–10%. Endoscopic therapy is indicated for actively bleeding Mallory-Weiss tears. **Mallory-Weiss tears are discussed in Chap. 334.**

ESOPHAGEAL VARICES The proportion of UGIB hospitalizations due to varices varies widely from ~2–40%, depending on the population. Patients with variceal hemorrhage have poorer outcomes than patients with other sources of UGIB. Esophageal varices are treated with endoscopic ligation and an IV vasoactive medication (octreotide, somatostatin, vapreotide, terlipressin) for 2–5 days. Combination of endoscopic and medical therapy is superior to either therapy alone in decreasing rebleeding. Over the long term, treatment with nonselective beta blockers plus endoscopic ligation is recommended because the combination is more effective than either alone in reduction of recurrent esophageal variceal bleeding. Transjugular intrahepatic portosystemic shunt (TIPS) is recommended in patients who have persistent or recurrent bleeding despite endoscopic and medical therapy. TIPS also is suggested for acute variceal bleeding in patients with advanced liver disease (Child-Pugh class B with score 8–9 and active bleeding at endoscopy, Child-Pugh class C with score 10–13), because randomized trials show significant decreases in rebleeding and mortality compared with standard endoscopic and medical therapy.

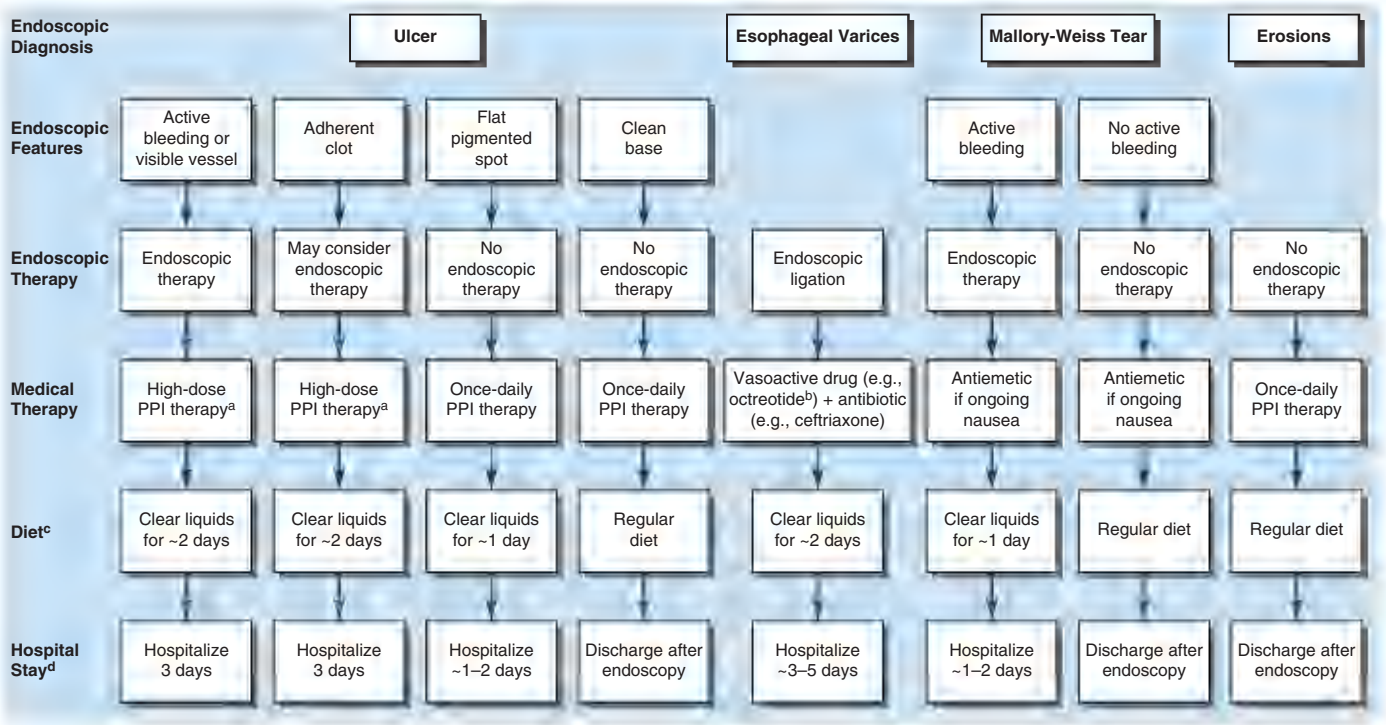

FIGURE 51-1 Suggested algorithm for patients with acute upper gastrointestinal bleeding (hematemesis, melena) based on endoscopic findings. [a]Intravenous bolus (80 mg) followed by infusion (8 mg/h) or by intermittent oral or intravenous doses (e.g., 40 mg 2–4 times per day) for 3 days. [b]Intravenous 50 μg bolus followed by 50 μg/h infusion for 2–5 days. [c]Diet after endoscopy, assuming no nausea or vomiting. [d]Duration after endoscopy assuming patient stable without further bleeding or concurrent medical conditions requiring hospitalization. PPI, proton pump inhibitor.

Portal hypertension is also responsible for bleeding from gastric varices, varices in the small and large intestine, and portal hypertensive gastropathy and enterocolopathy. Bleeding gastric varices are treated with endoscopic injection of tissue adhesive (e.g., *n*-butyl cyanoacrylate), TIPS, or retrograde transvenous obliteration.

EROSIVE DISEASE Erosions are endoscopically visualized breaks that are confined to the mucosa and do not cause major bleeding because arteries and veins are not present in the mucosa. Erosions in the esophagus, stomach, or duodenum commonly cause mild UGIB, with erosive esophagitis (primarily due to gastroesophageal reflux disease), gastritis, and duodenitis accounting for up to perhaps ~30% of UGIB hospitalizations. The most important cause of gastric and duodenal erosions is NSAID use: up to ~50% of patients who chronically ingest NSAIDs may have gastric erosions. Other potential causes of gastric erosions include alcohol intake, *H. pylori* infection, and stress-related mucosal injury.

Stress-related gastric mucosal injury occurs only in extremely ill patients, such as those with serious trauma, major surgery, burns covering more than one-third of the body surface area, major intracranial disease, or severe medical illness. Severe bleeding should not develop unless ulceration occurs. The mortality rate in these patients is high because of their serious underlying illnesses.

The incidence of bleeding from stress-related gastric mucosal injury has decreased dramatically in recent years, most likely due to better care of critically ill patients. Meta-analysis of randomized trials indicates prophylaxis with PPIs reduces clinically important GIB more than no prophylaxis or H_2-receptor antagonists without impacting mortality or risk of infection (e.g., pneumonia, *Clostridioides difficile*). A guideline suggested PPI prophylaxis in critically ill patients at high risk (≥4%) of bleeding, defined as mechanical ventilation without enteral nutrition, portal hypertension, cirrhosis, platelets $<50 \times 10^9$/L, international normalized ratio >1.5, or two of the following: mechanical ventilation with enteral nutrition, acute kidney injury, sepsis, or shock.

OTHER CAUSES Less common causes of UGIB include neoplasms, vascular ectasias (including hereditary hemorrhagic telangiectasias and gastric antral vascular ectasia), Dieulafoy's lesion (in which an aberrant vessel in the mucosa bleeds from a pinpoint mucosal defect), prolapse gastropathy (prolapse of proximal stomach into esophagus with retching), aortoenteric fistulas, and hemobilia or hemosuccus pancreaticus (bleeding from the bile duct or pancreatic duct).

Small-Intestinal Sources of Bleeding Patients without a source of GIB identified on upper endoscopy and colonoscopy were previously labeled as having obscure GIB. With the advent of improved diagnostic modalities, ~75% of GIB previously labeled obscure is now estimated to originate in the small intestine beyond the extent of a standard upper endoscopic exam. Small-intestinal GIB may account for ~5% of GIB cases. The most common causes in adults include vascular ectasias, neoplasm (e.g., gastrointestinal stromal tumor, carcinoid, adenocarcinoma, lymphoma, metastases), and NSAID-induced erosions and ulcers. Meckel's diverticulum is the most common cause of significant small-intestinal GIB in children, decreasing in frequency as a cause of bleeding with age. Other less common causes of small-intestinal GIB include Crohn's disease, infection, ischemia, vasculitis, small-bowel varices, diverticula, intussusception, Dieulafoy's lesions, aortoenteric fistulas, and duplication cysts.

Small-intestinal vascular ectasias are treated initially with endoscopic therapy, based on observational studies suggesting short-term efficacy. However, rebleeding is common, with pooled rebleeding rates of ~45% over a mean follow-up of ~2 years in systematic reviews, leading guidelines to suggest medical therapy if further bleeding occurs after endoscopic therapy. The best available evidence supports use of thalidomide, with a multicenter double-blind randomized trial showing marked reductions in bleeding episodes, transfusions, and hospitalizations. Monthly intramuscular injection of octreotide long-acting release also is suggested based on observational studies and a small open-label randomized trial. Other isolated lesions, such as tumors, generally require surgical resection.

Colonic Sources of Bleeding Hemorrhoids are probably the most common cause of lower GIB (LGIB); anal fissures also cause minor bleeding and pain. If these local anal processes, which rarely require hospitalization, are excluded, the most common cause of LGIB in adults is diverticulosis. Other causes include vascular ectasias (especially in the proximal colon of patients >70 years), neoplasms (primarily adenocarcinoma), colitis (ischemic, infectious, Crohn's or ulcerative colitis, NSAID-induced colitis or ulcers), postpolypectomy bleeding, and radiation proctopathy. Rarer causes include solitary rectal ulcer syndrome, varices (most commonly rectal), lymphoid nodular hyperplasia, vasculitis, trauma, and aortocolic fistulas. In children and adolescents, the most common colonic causes of significant GIB are inflammatory bowel disease and juvenile polyps.

Diverticular bleeding is abrupt in onset, usually painless, sometimes massive, and often from the right colon; chronic or occult bleeding is not characteristic. Case series from the United States and Europe suggest colonic diverticula stop bleeding spontaneously in ≥90% of patients, with rebleeding on long-term follow-up as low as ~15% over 4–5 years. Rebleeding is substantially higher in reports from Asia. Case series suggest endoscopic therapy may decrease recurrent bleeding in the uncommon case when colonoscopy identifies the specific bleeding diverticulum. When diverticular bleeding is found at angiography, transcatheter arterial embolization by superselective technique prevents further bleeding in most patients. Segmental surgical resection is recommended for refractory diverticular bleeding.

Bleeding from colonic vascular ectasias may be overt or occult; it tends to be chronic and only occasionally hemodynamically significant. Endoscopic hemostatic therapy may be used in the treatment of vascular ectasias, as well as discrete bleeding ulcers and postpolypectomy bleeding. Transcatheter arterial embolization also may be attempted for persistent bleeding from vascular ectasias and other lesions, although rebleeding is higher in nondiverticular LGIB at ~45%. Surgical therapy is generally required for major persistent or recurrent bleeding from colonic sources that cannot be treated medically, endoscopically, or angiographically. Patients with Heyde's syndrome (bleeding vascular ectasias and aortic stenosis) appear to benefit from aortic valve replacement.

APPROACH TO THE PATIENT

Gastrointestinal Bleeding

INITIAL ASSESSMENT

Measurement of the heart rate and blood pressure is the best way to initially assess a patient with GIB. Clinically significant bleeding leads to postural changes in heart rate or blood pressure, tachycardia, and, finally, recumbent hypotension. In contrast, hemoglobin does not fall immediately with acute GIB, due to proportionate reductions in plasma and red cell volumes ("people bleed whole blood"). Thus, hemoglobin may be normal or only minimally decreased at initial presentation of a severe bleeding episode. As extravascular fluid enters the vascular space to restore volume, the hemoglobin falls, but this process may take up to 72 h. Transfusion is recommended when the hemoglobin drops below 7 g/dL, based on a large randomized trial showing this restrictive transfusion strategy decreases rebleeding and death in acute UGIB compared with a transfusion threshold of 9 g/dL. Patients with slow, chronic GIB may have very low hemoglobin values despite normal blood pressure and heart rate. With the development of iron-deficiency anemia, the mean corpuscular volume is low and red-blood-cell distribution width is increased.

DIFFERENTIATION OF UGIB FROM LGIB

Hematemesis indicates an UGIB source. Melena indicates blood has been present in the gastrointestinal (GI) tract for ≥14 h and as long as 3–5 days. The more proximal the bleeding site, the more likely melena will occur. Hematochezia usually represents a lower GI source of bleeding, although an upper GI lesion may bleed so briskly that blood transits the bowel before melena develops. When

hematochezia is the presenting symptom of UGIB, it is associated with hemodynamic instability and dropping hemoglobin. Bleeding lesions of the small bowel may present as melena or hematochezia. Other clues to UGIB include hyperactive bowel sounds and elevated blood urea nitrogen (due to volume depletion and blood proteins absorbed in the small intestine).

A nonbloody nasogastric aspirate may be seen in ~15% of patients with UGIB who present with clinically serious hematochezia. A bile-stained appearance does not exclude UGIB because reports of bile in the aspirate are incorrect in ~50% of cases. Testing of aspirates that are not grossly bloody for occult blood is not useful.

EVALUATION AND MANAGEMENT OF UGIB (FIG. 51-1)

Initial Risk Assessment Baseline characteristics predictive of rebleeding and death include hemodynamic compromise (tachycardia or hypotension), increasing age, and comorbidities. Risk assessment tools may be used to identify patients with very low risk. Discharge from the emergency room with outpatient management is suggested for patients with a Glasgow-Blatchford score (possible range 0–23, **Table 51-1**) of 0–1 because only ~1% of patients who require transfusion, require hemostatic intervention, or die have a score of 0–1.

Pre-Endoscopic Medications PPI infusion may be considered at presentation; it modestly decreases need for endoscopic therapy due to a reduction in high-risk ulcer stigmata (e.g., active bleeding) but does not improve clinical outcomes such as further bleeding or death. The promotility agent erythromycin, 250 mg intravenously ~30–90 min before endoscopy, is suggested to improve visualization at endoscopy, thereby reducing the need for repeat endoscopy and hospital stay. Cirrhotic patients presenting with UGIB should be given an antibiotic (e.g., ceftriaxone) and IV vasoactive medication (e.g., octreotide) upon presentation. Antibiotics decrease bacterial infections, rebleeding, and mortality, and vasoactive medications improve control of bleeding in the 12 h after presentation.

Endoscopy Upper endoscopy should be performed within 24 h in most patients hospitalized with UGIB whether they have clinical features predicting low or high risk of further bleeding and death. Even in high-risk patients, more urgent endoscopy (performed within 6–12 h of gastroenterology consultation) does not improve clinical outcomes, and some observational studies suggest increased mortality with endoscopy within 6–12 h in high-risk patients. Early endoscopy in low-risk patients (e.g., hemodynamically stable without severe comorbidities) identifies low-risk findings (e.g., clean-based ulcers, erosions, nonbleeding Mallory-Weiss tears) that allow discharge in ≥40% of patients, thereby reducing hospital stay

TABLE 51-1 Glasgow-Blatchford Score

RISK FACTORS AT ADMISSION	SCORE
Blood urea nitrogen (mg/dL)	
18.2 to <22.4	2
22.4 to <28.0	3
28.0 to <70.0	4
≥70.0	6
Hemoglobin (g/dL)	
12.0 to <13.0 (men); 10.0 to <12.0 (women)	1
10.0 to <12.0 (men)	3
<10.0	6
Systolic blood pressure (mmHg)	
100–109	1
90–99	2
<90	3
Heart rate (beats per minute)	
≥100	1
Melena	1
Syncope	2
Hepatic disease	2
Cardiac failure	2

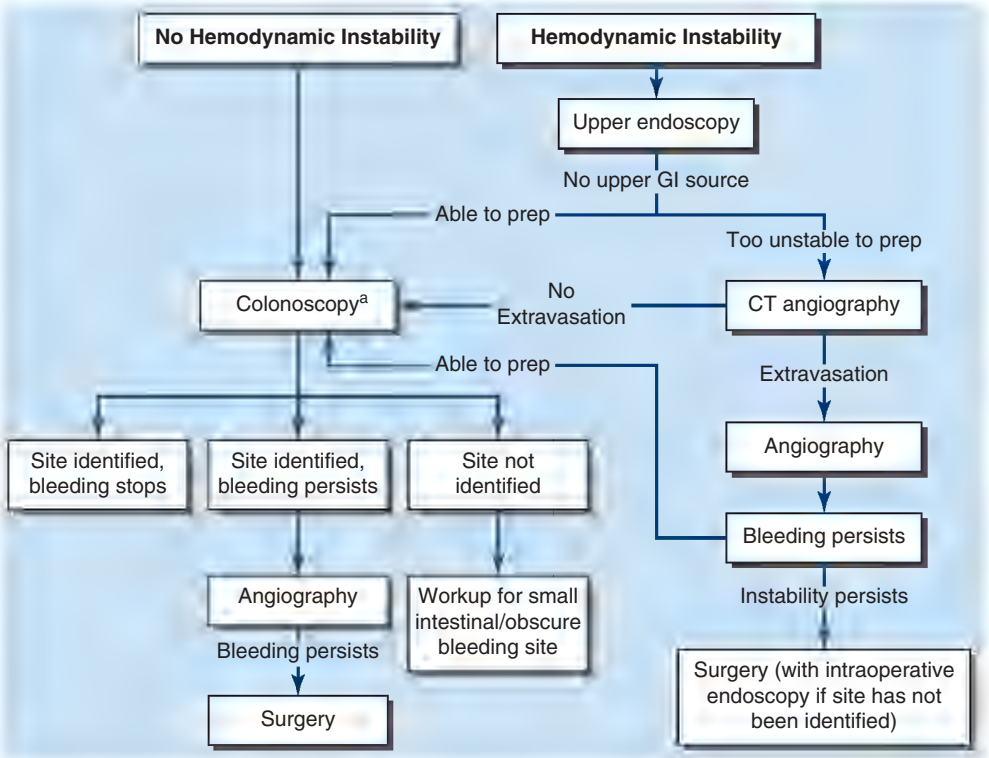

FIGURE 51-2 Suggested algorithm for patients with acute lower gastrointestinal (GI) bleeding (hematochezia). [a]For patients <35 years old with minor bleeding (e.g., blood on toilet paper), normal blood pressure and heart rate, normal hemoglobin and iron panel, and no family history of colorectal cancer, some would suggest flexible sigmoidoscopy is adequate.

and costs. Patients with high-risk endoscopic findings (e.g., varices, ulcers with active bleeding or a visible vessel) benefit from hemostatic therapy at endoscopy.

EVALUATION AND MANAGEMENT OF LGIB (FIG. 51-2)

Patients with hematochezia and hemodynamic instability should have upper endoscopy to rule out an upper GI source before evaluation of the lower GI tract.

Colonoscopy after an oral lavage solution is the procedure of choice in most patients admitted with LGIB. If ongoing hemodynamically significant hematochezia precludes bowel preparation and colonoscopy, computed tomography (CT) angiography is suggested. If extravasation is seen, angiography is performed, allowing treatment with transcatheter arterial embolization. If no extravasation is seen, colonoscopy is subsequently performed.

EVALUATION AND MANAGEMENT OF SMALL-INTESTINAL OR OBSCURE GIB

In patients with severe bleeding suspected to be from the small intestine, CT angiography or angiography is the initial test. For others, video capsule endoscopy is generally the next diagnostic test suggested, although repeat upper and lower endoscopy may be considered because second-look procedures in some observational studies identify a source in up to ~25% of cases (a push enteroscopy, usually performed with a pediatric colonoscope to inspect the entire duodenum and proximal jejunum, may be substituted for a repeat standard upper endoscopy). Systematic reviews report a diagnostic yield with capsule endoscopy of ~55%. Limitations of capsule endoscopy include the inability to fully visualize the small intestinal mucosa, sample tissue, or apply therapy.

If capsule endoscopy is positive, management is dictated by the finding. If capsule endoscopy is negative, clinically stable patients may be observed and treated with iron if iron deficiency is present. In those with further bleeding (e.g., need for transfusions), additional testing is performed. A second capsule endoscopy may be considered because some observational studies report it identifies a source in up to 50% or more of cases. CT enterography also may follow a negative capsule endoscopy, given its higher sensitivity for small-intestinal masses (CT enterography also may be used initially instead of video capsule in patients with concern for small bowel narrowing [e.g., stricture, prior surgery or radiation, Crohn's disease]).

"Deep" enteroscopy (double-balloon, single-balloon, or spiral enteroscopy) is commonly the next test after capsule endoscopy for clinically important GIB documented or suspected to be from the small intestine because it allows the endoscopist to examine, obtain specimens from, and provide therapy to much or all of the small intestine. Other imaging techniques sometimes used in evaluation of obscure GIB include CT angiography, angiography, ^{99m}Tc-labeled red-blood-cell scintigraphy, and ^{99m}Tc-pertechnetate scintigraphy for Meckel's diverticulum (especially in young patients). If all tests are unrevealing, intraoperative endoscopy is indicated in patients with severe recurrent or persistent bleeding requiring repeated transfusions.

POSITIVE FECAL OCCULT BLOOD TEST

Fecal occult blood testing is recommended only for colorectal cancer screening, beginning at age 45 years in average-risk adults. A positive test necessitates colonoscopy. If evaluation of the colon is negative, further workup is not recommended unless iron-deficiency anemia or GI symptoms are present.

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52 Jaundice

Savio John, Daniel S. Pratt

Jaundice is a yellowish discoloration of body tissues resulting from the deposition of bilirubin. Tissue deposition of bilirubin occurs only in the presence of serum hyperbilirubinemia and is a sign of either liver disease or, less often, a hemolytic disorder or disorder of bilirubin metabolism. The degree of serum bilirubin elevation can be estimated by physical examination. Slight increases in serum bilirubin level are best detected by examining the sclerae for icterus. Sclerae have a particular affinity for bilirubin due to their high elastin content, and the presence of scleral icterus indicates a serum bilirubin level of at least 51 μmol/L (3 mg/dL). The ability to detect scleral icterus is made more difficult if the examining room has fluorescent lighting. If the examiner suspects scleral icterus, a second site to examine is underneath the tongue. As serum bilirubin levels rise, the skin will eventually become yellow in light-skinned patients and even green if the process is long-standing; the green color is produced by oxidation of bilirubin to biliverdin.

The differential diagnosis for yellowing of the skin is limited. In addition to jaundice, it includes carotenoderma; the use of drugs including quinacrine, sunitinib, and sorafenib; and excessive exposure to phenols. Carotenoderma, a yellow coloring of the skin, is associated with diabetes, hypothyroidism, and anorexia nervosa, but most commonly, it is caused by the ingestion of an excessive amount of vegetables and fruits such as carrots, leafy vegetables, squash, peaches, and oranges that contain carotene. In jaundice, the yellow coloration of the skin is uniformly distributed over the body, whereas in carotenoderma, the pigment is concentrated on the palms, soles, forehead, and nasolabial folds. Carotenoderma can be distinguished from jaundice by the sparing of the sclerae. Quinacrine causes a yellow discoloration of the skin in 4–37% of patients treated with it. It has also been reported with the use of the tyrosine kinase inhibitors sunitinib and sorafenib.

Another sensitive indicator of increased serum bilirubin is darkening of the urine, which is due to the renal excretion of conjugated bilirubin. Patients often describe their urine as tea- or cola-colored. Bilirubinuria indicates an elevation of the direct serum bilirubin fraction and, therefore, the presence of liver or biliary disease.

Serum bilirubin levels increase when an imbalance exists between bilirubin production and clearance. A logical evaluation of the patient who is jaundiced requires an understanding of bilirubin production and metabolism.

PRODUCTION AND METABOLISM OF BILIRUBIN

(See Chap. 349) Bilirubin, a tetrapyrrole pigment, is a breakdown product of heme (ferroprotoporphyrin IX). About 80–85% of the